

Skill Arena - Phase 5 Document 3

Game Engine Architecture, Matchmaking Engine, Challenge Generation & Gameplay Processing Framework

Document Purpose

Define the core gameplay architecture responsible for challenge generation, matchmaking, gameplay execution and competitive fairness.

Business Objective

Deliver infinitely scalable skill-based gameplay with fair matchmaking and server-authoritative processing.

Game Engine Philosophy

Every challenge must be unique, verifiable, scalable and resistant to exploitation.

Core Components

Game Engine, Matchmaking Engine, Challenge Generator, Difficulty Engine, Validation Engine and Replay Engine.

Game Engine Responsibilities

Challenge creation, state management, rule enforcement, progression tracking and outcome validation.

Maze Challenge Generator

Procedurally generates unique challenge layouts using controlled randomization.

Challenge Uniqueness Controls

Prevent repeated patterns, duplicate layouts and predictable challenge structures.

Challenge Difficulty Engine

Adjusts complexity based on category, progression, league and challenge tier.

Infinite Progression Support

No hard cap on progression. Difficulty scales dynamically as players advance.

Challenge Metadata

Challenge ID, Seed, Difficulty Score, Category, Validation Hash and Replay Reference.

Matchmaking Engine

Matches players using skill, progression, reputation and competitive fairness metrics.

Matchmaking Factors

ELO Rating, XP Level, Difficulty Completed, House Reputation, League and Challenge History.

Cross-Tier Matchmaking

Players may voluntarily challenge higher categories with risk disclosures.

Queue Management

Queue creation, queue balancing, wait-time optimization and match allocation.

Gameplay Processing Engine

Processes player actions, state updates, validation checks and match events.

Server Authoritative Processing

Gameplay state controlled exclusively by trusted backend services.

Real-Time Synchronization

Synchronizes challenge progress, timers and match status across participants.

House Challenge Processing

Manages challenge eligibility, house rules, payout logic and progression requirements.

Replay Integration

All matches and challenges linked to replay generation and verification services.

Reward Integration

Communicates with wallet and treasury systems after validated completion.

Scalability Requirements

Support millions of concurrent challenges and large-scale tournaments.

Fault Tolerance

Challenge recovery, queue recovery and gameplay state resilience.

Monitoring Requirements

Track challenge generation, matchmaking health and gameplay performance.

Security Requirements

Challenge seeds, validation logic and reward calculations protected server-side.

Audit Requirements

All challenge creation, matchmaking decisions and outcomes recorded permanently.

Acceptance Criteria

Gameplay remains fair, scalable, verifiable and resistant to manipulation.